SUPPORTING INFORMATION

Modulating responses of toehold switches by an inhibitory hairpin

Soo-Jung Kim, Matthew Leong, Matthew B. Amrofell, Young Je Lee, and Tae Seok Moon

METHODS

Strain and culture media

The *E. coli* DH10B strain or the BL21 Star (DE3) strain was used for all experiments. Cells were grown in LB media (5 g/L yeast extract, 10 g/L tryptone, and 10 g/L NaCl) supplemented with the appropriate antibiotics: ampicillin (100 µg/mL) and kanamycin (20 µg/mL). Inducers were used at the following concentrations: IPTG (isopropyl β-D-1-thiogalactopyranoside, 0–20 mM) and 3OC6 (3-oxohexanoyl-homoserine lactone, 0–5 µM). All chemical reagents used in this study were purchased from Sigma-Aldrich (St. Louis, MO).

Design of toehold switch RNAs and trigger RNAs

Two types of toehold switches in Figure 1 (Sw-G5-G3n*) and Figure 2 (Sw-T0-T3) were designed as follows. The one-hairpin switch RNA (Sw-G3n5) was derived from the previously reported switch RNA with modification[1]. To improve RNA stability, additional hairpin sequence (aacgggctgcaataacgcagcccaat; shown as its DNA sequence) was inserted upstream of the switch RNA, and the 21-nt linker region was placed between the switch RNA and *gfpmut3* gene (Supplementary Figure S1A). The trigger RNA was restricted to total 30 nucleotides for binding to the switch RNA. For the complete trigger transcript, a stabilizing 5'-hairpin sequence (aaagtcaaggatagcgccttgacgac) and the his[min](S) terminator were included upstream and downstream of the 30-nt sequence, respectively. To construct the two-hairpin switch RNA (Sw-G5-G3n* where * = 5, 8, or 11; Supplementary Figure S1B), an inhibitory hairpin (G5) was placed upstream of the main switch hairpin (G3n*).

The second type of toehold switches (Sw-T0-T3) was designed (Supplementary Figure S1C) by using the NUPACK software[2] (http://www.nupack.org/design/new). Our design is shown below (using DU+ notation):

trials = 10
structure switch = U18 D6(U3 D6(U12) U3) U6 D11 (U1 D6 (U15) U1) U50
structure trigger = U30
domain a = N21
domain b =
AUACAGAAACAGAGGAGAUAUAGAAUGAGACAAUGGaaccuggcggcagcgcaaaagaugagua
aaggagaagaacuuuucacugg
domain parttwotrig = N15
domain partonetrig = N15

prevent = aaaa, cccc, gggg, uuuu, kkkkkk, mmmmmm, rrrrrr, ssssss, wwwwww, yyyyyy
switch.seq = partonetrig parttwotrig a partonetrig parttwotrig b
trigger.seq = parttwotrig* partonetrig*

'domain a' is a sequence of the inhibitory hairpin (T0), and 'domain b' is a sequence of the main switch RNA (T3: part of hairpin in grey; 21-nt linker in pink; and part of *gfpmut3* in green). The following sequences were prevented: aaaa, cccc, gggg, uuuu, kkkkkk, mmmmmm, rrrrrr, ssssss, wwwwww, and yyyyyy. All RNA regulator sequences used in this study are shown in Supplementary Table 2 (shown as their DNA sequences).

Construction of toehold switch plasmids

All DNA sequences for toehold switch RNAs were synthesized using oligonucleotides and Phusion High-fidelity DNA polymerase (New England Biolabs, Ipswich, MA) through overlap extension PCR[3]. The amplified and purified DNA fragments were then assembled using Gibson assembly method[4]. All DNA sequences for trigger RNAs were inserted into plasmids using oligonucleotides using inverse PCR, and the amplified and purified DNA fragments were simultaneously phosphorylated using T4 polynucleotide kinase (New England Biolabs, Ipswich, MA) and ligated using T4 DNA ligase (New England Biolabs, Ipswich, MA) at room temperature for 1 h. The assembled products were transformed into electrocompetent *E. coli* DH10B using an electroporator (Eppendorf Eporator, Fisher scientific, St. Peters, MO). The transformants were grown on LB agar with appropriate antibiotics at 37°C overnight. Plasmids were purified from the overnight culture using a miniprep kit (Zymo Research, Irvine, CA), and DNA sequences of the constructed plasmids were verified by DNA sequencing. All oligonucleotides were purchased from Integrated DNA Technologies (IDT, Coralville, IA). Plasmids used in this study and genetic parts including RNA regulator sequences are shown in Supplementary Tables S1 and S2, respectively. Detailed schematics of the representative toehold switches constructed in this study (with length information), which are based on secondary RNA structures designed and predicted by NUPACK[2], are also shown in Supplementary Figure S1.

Fluorescence and absorbance measurements

Cells were grown overnight in 5 mL LB media with appropriate antibiotics at 37°C and 250 rpm in a New Brunswick Excella E25 shaking incubator. The overnight cultures (1% v/v) were subcultured in fresh 5 mL LB media with appropriate antibiotics and grown for 2 h at 37°C and 250 rpm. The cultured cells were transferred to 0.6 mL of fresh LB media with appropriate antibiotics and inducers at concentrations as indicated in figure captions in deep 96-well plates (Eppendorf). After cultivation for 8 h at 37°C and 250 rpm, the cells were centrifuged and the cell pellet was resuspended in 0.2 mL PBS (Phosphate-buffered saline, pH 8.0). The resuspended cells were transferred in 96-well microplate, and cell densities and fluorescence levels were measured using a Tecan microplate reader (Infinite M200 Pro). The output (F_{out}) was reported by calculating $F_{out} = [(F_{experimental}/Abs_{experimental}) - (F_{negative\ control}/Abs_{negative\ control})]$, where the negative control was the DH10B strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm.

Data fitting

To obtain the relative promoter activity for TrT3 in Figure 2B (for DH10B), the Hill equation was used with some modifications[5]. The transfer function for the Lac promoter (by which TrT3 is transcribed) is described by the general Hill equation:

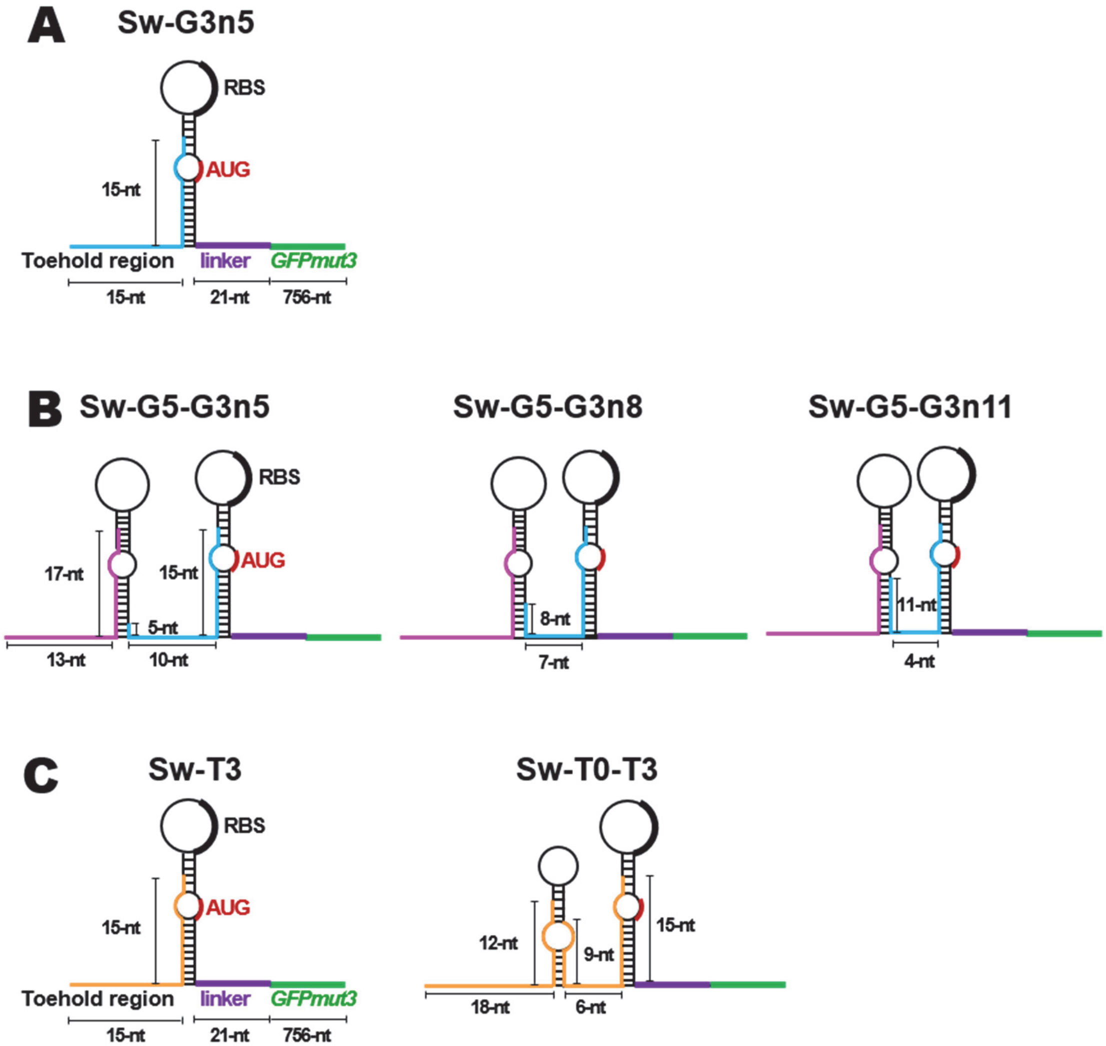
$$P_x = (F_{max} - F_{min})\frac{I^{n_x}}{I^{n_x} + K_x^{n_x}} + F_{min} \text{ [Equation 1]}$$

where P_x is the relative promoter activity for TrT3 (normalized by the maximum fluorescence value obtained by the Lac promoter), I is the IPTG concentration, n_x is the Hill coefficient, and K_x is the half-maximal concentration. F_{max} and F_{min} are the maximal and basal normalized fluorescence, respectively. The Hill equation was fitted to the data by minimizing the root mean square error (Supplementary Figure S3). The fitted parameters are: $n_x = 0.57$, $K_x = 0.050$ mM, $F_{max} = 1.031$, and $F_{min} = 0.069$. Similarly, the fitted parameters for the other Lac promoter (used in BL21 Star (DE3)) and the Lux promoter are listed in the corresponding figures.

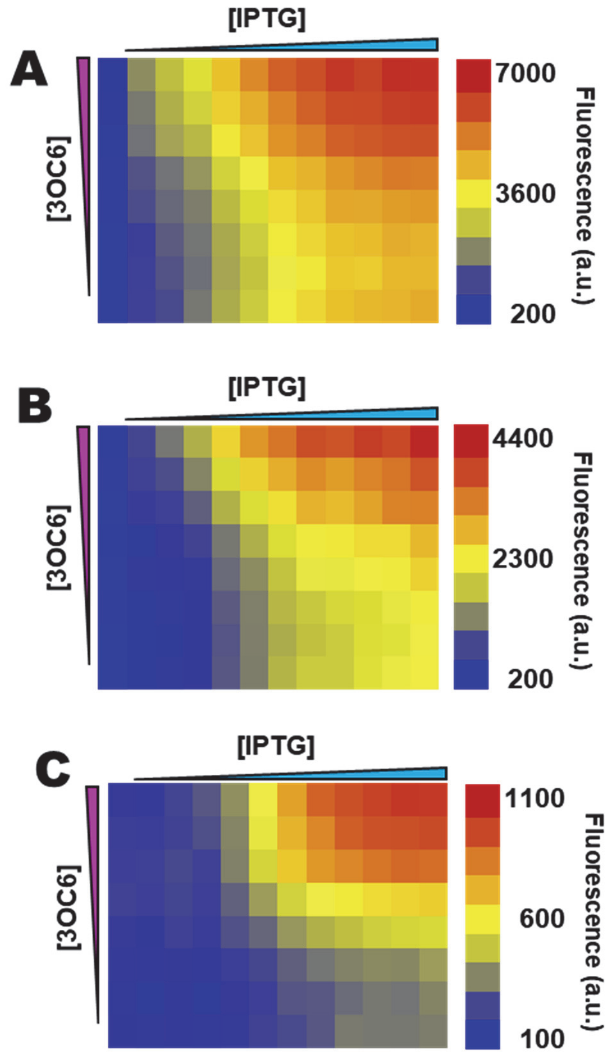
For ultrasensitive switches, F_{out} can be described by $F_{out} = a[P_x^{nH} / (P_x^{nH} + K_H^{nH})] + b$, and the normalized output ($F_{norm} = (F_{out} - F_{min})/(F_{max} - F_{min})$) can be described by the non-dimensionalized Hill-type equation:

$$F_{norm} = A\frac{{P_x}^{n_H}}{{P_x}^{n_H} + K_H^{n_H}} + B \text{ [Equation 2]}$$

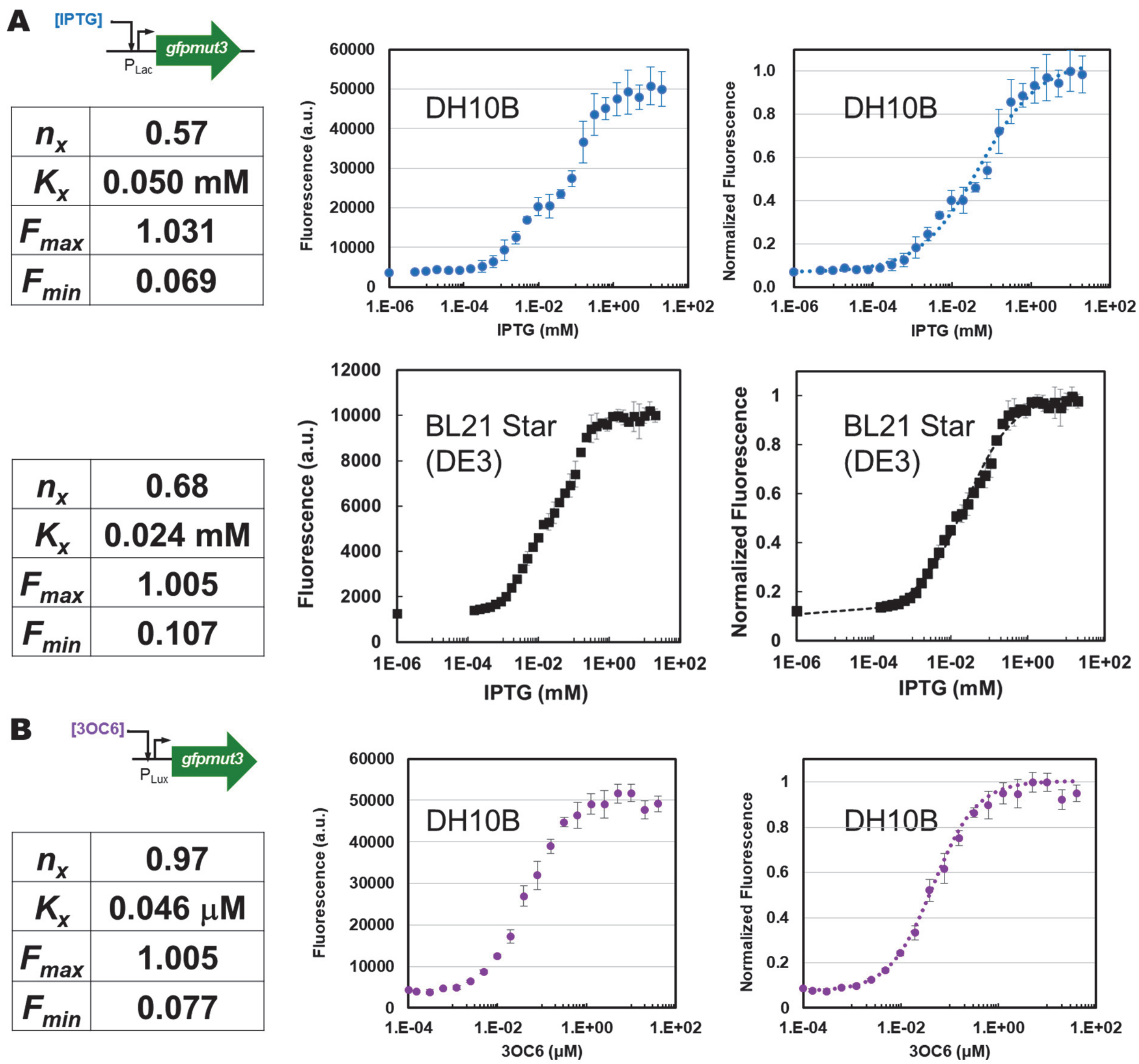
where P_x is the relative promoter activity for TrT3 (for Figure 2B) from Equation 1, n_H is the apparent Hill coefficient, and K_H is the apparent half-maximal promoter activity. The non-dimensionalized parameters are: $A=a/(F_{max} - F_{min})$ and $B=(b - F_{min})/(F_{max} - F_{min})$ where F_{min} and F_{max} are the minimum and maximum fluorescence outputs experimentally determined, respectively. Alternatively, F_{min} and F_{max} can be determined by fitting $F_{out} = (F_{max} - F_{min})[P_x^{nH} / (P_x^{nH} + K_H^{nH})] + F_{min}$. When comparing among switches with the same number of hairpins and similar minimum fluorescence outputs (Figures 2C and S5), we used the latter, while the former (i.e., the non-dimensionalized equation) was used for easy graphical comparison when switches had very different minimum fluorescence outputs with different numbers of hairpins (Figures 2B, 2D, S4, and S6). The equation was fitted to the data by minimizing the root mean square error. For all cases, R^2 values were greater than 0.95. In one-hairpin switch RNA, the fitted parameters (Figure 2B) are: $n_H = 2.8$, $K_H = 0.64$, $A = 1.20$, and $B = 0.02$ (with experimental data: $F_{min} = 296$ a.u. and $F_{max} = 5726$ a.u.). In two-hairpin switch RNA, the fitted parameters (Figure 2B) are: $n_H = 7.0$, $K_H = 0.76$, $A = 1.10$, and $B = 0.04$ (with experimental data: $F_{min} = 61$ a.u. and $F_{max} = 1367$ a.u). Similarly, the fitted parameters for the other toehold switches are listed in the corresponding figures.



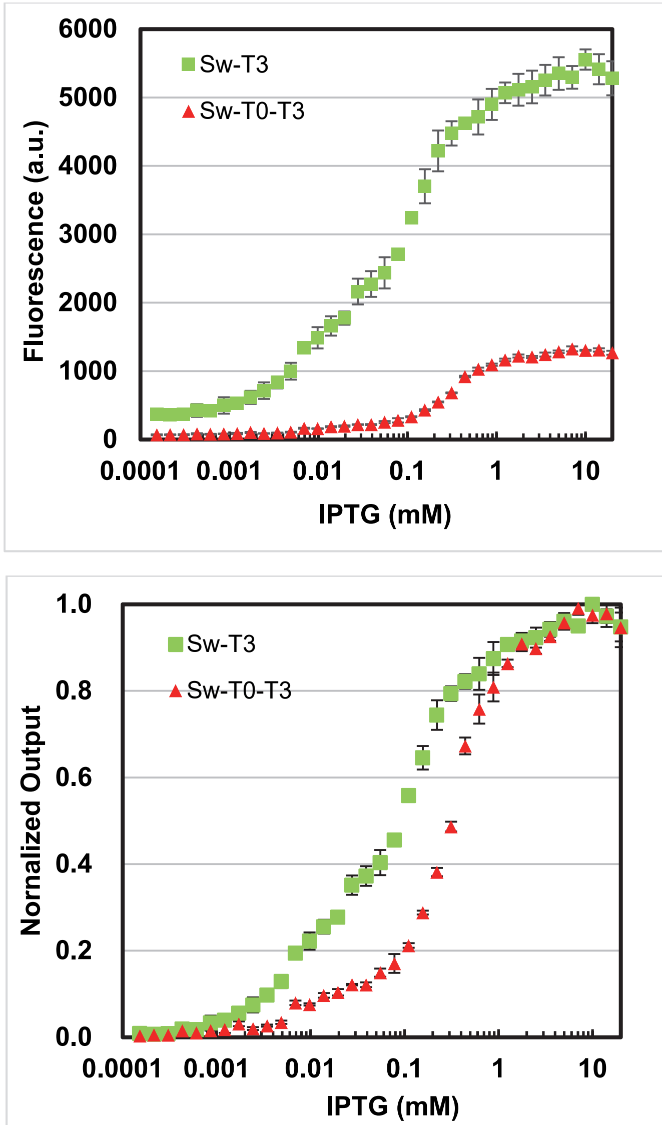
Supplementary Figure S1. Detailed schematics of the representative toehold switches constructed in this study. Design specifics for the one-hairpin switch RNA (A), the two-hairpin switch RNA / two-trigger RNA system (B), and the two-hairpin switch RNA / one-trigger RNA system together with its one-hairpin switch RNA control (C). Binding sites of triggers (TrG3n*, TrG5, and TrT3 where * is 5, 8, or 11) are in sky blue, pink, and orange, respectively.



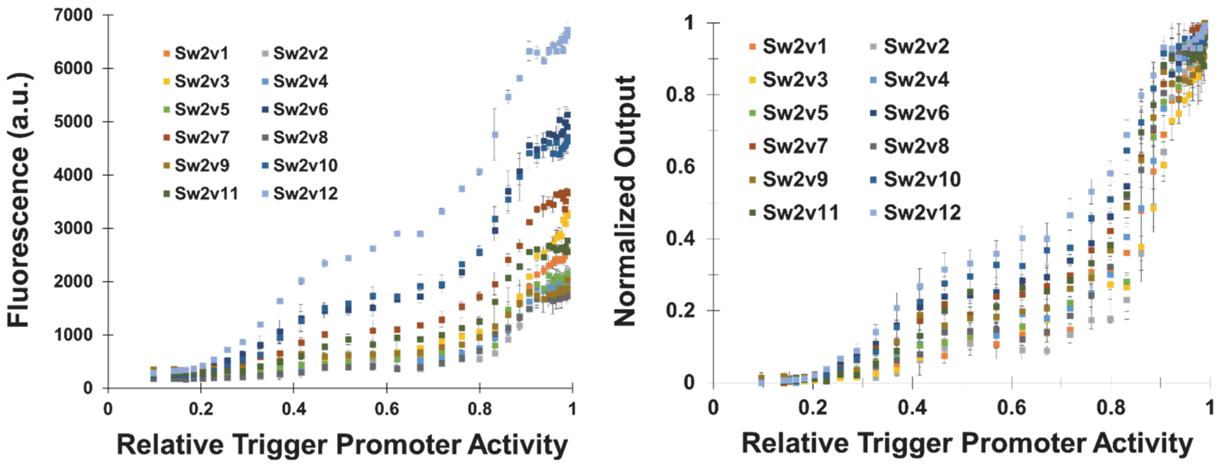
Supplementary Figure S2. Fluorescence data from the two-hairpin switch RNA (Sw-G5-G3n*) and two-trigger RNA (TrG5 and TrG3n*) system presented in Figure 1D. Heatmaps showing the fluorescence levels of Sw-G5-G3n5 (A), Sw-G5-G3n8 (B), and Sw-G5-G3n11 (C). The fluorescence (a.u.) is reported by calculating $[(F_{experimental}/Abs_{experimental}) - (F_{negative\ control}/Abs_{negative\ control})]$, where the negative control is the DH10B strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm. Interestingly, the maximum fluorescence (a.u.) decreased with the decreasing ‘a’ length, reducing the dynamic range. The experiments were performed at IPTG concentrations of 0, 0.01, 0.02, 0.04, 0.08, 0.16, 0.31, 0.63, 1.25, 2.5, 5 and 10 mM (from left to right) to transcribe TrG3n5 (A), TrG3n8 (B), or TrG3n11 (C); and at 3OC6 concentrations of 0, 0.005, 0.01, 0.025, 0.05, 0.25, 1 and 5 μM (from bottom to top) for TrG5 expression. Experimental data are the average of three biological replicates.



Supplementary Figure S3. Transfer functions of the promoters used for trigger RNA expression. (A) The Lac promoter transfer functions in DH10B and BL21 Star (DE3) at different IPTG concentrations (0–20 mM). The his [min] (S) and sgl Term terminators were used for DH10B and BL21 Star (DE3), respectively. **(B)** The Lux promoter transfer functions in DH10B at different 3OC6 concentrations (0–40 μM). Normalized fluorescence values were obtained by dividing the fluorescence output values (calculated using $[(F_{experimental}/Abs_{experimental}) - (F_{negative\ control}/Abs_{negative\ control})]$ as described in the Methods section) by the maximum value. The dotted line represents the regression line to fit the Hill equation described in Methods (Equation 1): $(F_{max} - F_{min})[I^{n_x}/(I^{n_x} + K_x^{n_x})] + F_{min}$, where I is the inducer concentration. The fitted parameters are shown in the figure. The error bars represent the standard deviation of the values from three biological replicates.

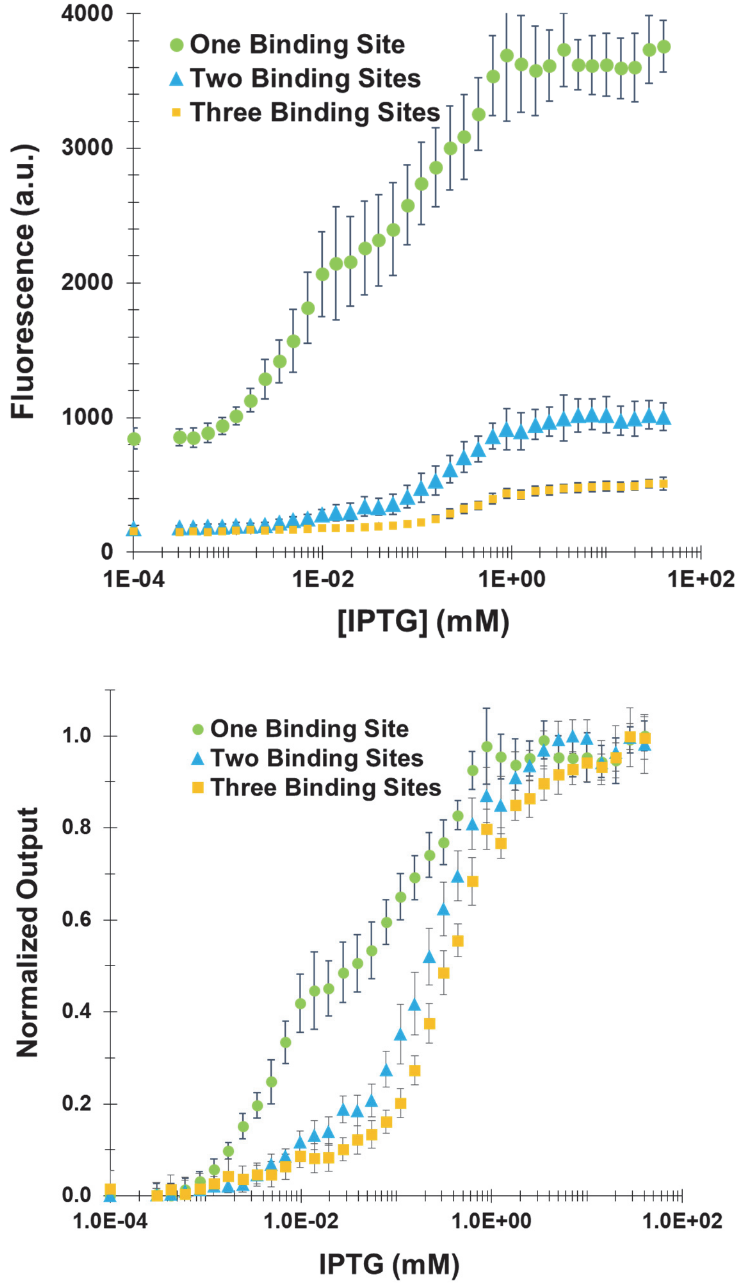


Supplementary Figure S4. The fluorescence data (Figure 2B) without (top) and with (bottom) normalization as a function of the IPTG concentration. The data are shown with various concentrations of IPTG (0–20 mM) to express TrT3. For the top figure, the fold changes were 19X (T3) and 22X (T0-T3). The fluorescence (a.u.) is reported by calculating [($F_{experimental}/Abs_{experimental}$) – ($F_{negative\ control}/Abs_{negative\ control}$)], where the negative control is the DH10B strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm. See Figure 2 for the genetic circuit and normalized fluorescence data as a function of the trigger RNA promoter activity. The error bars represent the standard deviation of the values from three biological replicates.

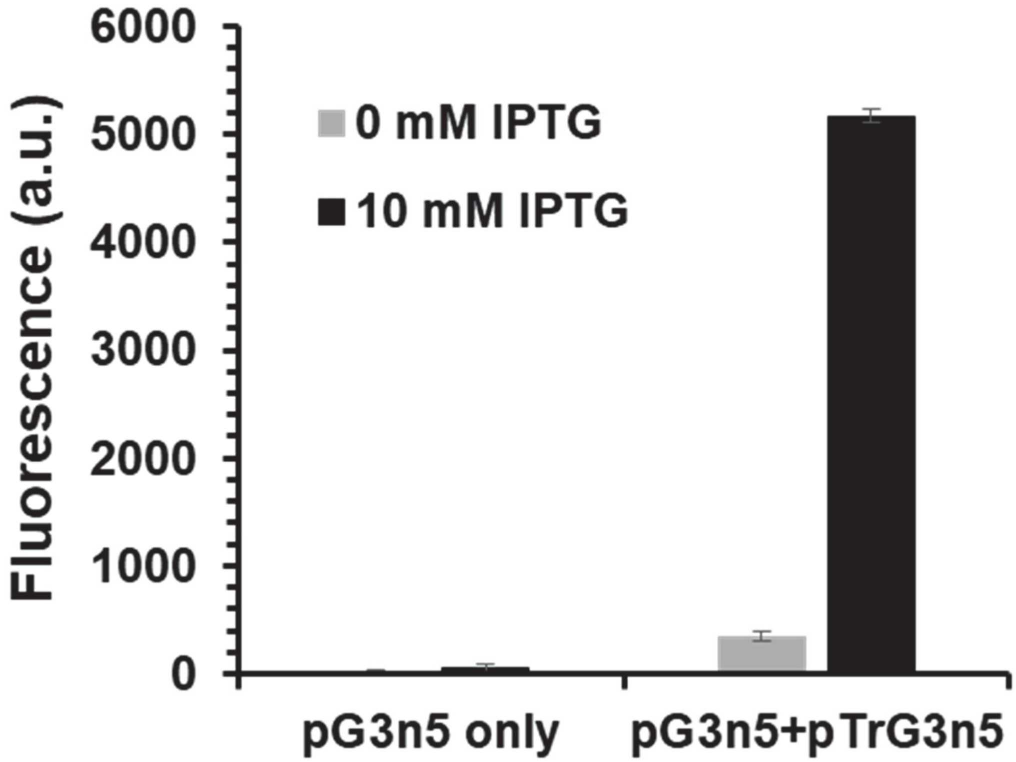


Switch	Hairpin-to-Hairpin Spacing (nt)	Fmin (au)	Fmax (au)	n	K	Fold Change	R^2
Sw2v1	3	255	2596	7.2	0.76	10.2	0.994
Sw2v2	3	208	2208	9.5	0.81	10.6	0.991
Sw2v3	2	213	3288	6.9	0.79	15.5	0.987
Sw2v4	2	181	2055	7.4	0.74	11.4	0.989
Sw2v5	5	237	2085	6.7	0.73	8.8	0.988
Sw2v6	7	282	5131	4.4	0.63	18.2	0.987
Sw2v7	9	256	3654	5.3	0.67	14.3	0.985
Sw2v8	2	174	1744	7.9	0.73	10.0	0.993
Sw2v9	3	347	2034	5.2	0.70	5.9	0.988
Sw2v10	5	236	4636	4.1	0.59	19.6	0.989
Sw2v11	7	262	2768	5.0	0.67	10.6	0.987
Sw2v12	9	286	6721	3.7	0.55	23.5	0.991

Supplementary Figure S5. The effect of the hairpin-to-hairpin spacing on the toehold switch output in the BL21 Star (DE3) strains. The fluorescence (a.u.) is reported by calculating $[(F_{experimental}/Abs_{experimental}) - (F_{negative\ control}/Abs_{negative\ control})]$, where the negative control is the BL21 Star (DE3) strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm. The normalized output $((F_{out} - F_{min})/(F_{max} - F_{min}))$ is also shown here. The experiment was performed at 36 different IPTG concentrations (0–20 mM). Data were fit to the Hill-type equation: $F_{out} = (F_{max} - F_{min})[P_x^n / (P_x^n + K^n)] + F_{min}$, where P_x is the relative promoter activity for trigger RNA (expressed by the Lac promoter in BL21 Star (DE3)). See Table S2-2 for genetic part sequences and Figure S3 for the Lac promoter transfer function. The fold change was also calculated by the ratio (F_{max}/F_{min}). The error bars represent the standard deviation of the values from three replicates.



Supplementary Figure S6. The fluorescence data (Figure 2D) without (top) and with (bottom) normalization as a function of the IPTG concentration. The data are shown with 36 concentrations of IPTG (0–40 mM) to express the trigger RNA (TrVKFT3) in BL21 Star (DE3). For the top figure, the fold changes were 4.4X (one site), 5.5X (two sites), and 3.3X (three sites). The fluorescence (a.u.) is reported by calculating $[(F_{experimental}/Abs_{experimental}) - (F_{negative\ control}/Abs_{negative\ control})]$, where the negative control is the BL21 Star (DE3) strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm. See Figure 2 for the normalized fluorescence data as a function of the trigger RNA promoter activity. The error bars represent the standard deviation of the values from three biological replicates.



Supplementary Figure S7. The effect of the leaky input promoter on the dynamic range. pG3n5+pTrG3n5, the same switch as that of Figure 1A or 1B. pG3n5 only, an OFF control switch (G3n5 fused to *gfpmut3*) without the trigger RNA (TrG3n5). The fluorescence (a.u.) is reported by calculating [($F_{experimental}$/$Abs_{experimental}$) – ($F_{negative\ control}$/$Abs_{negative\ control}$)], where the negative control is the DH10B strain without a plasmid. F is the measured fluorescence (excitation at 483 nm and emission at 530 nm), and Abs is the measured absorbance at 600 nm. The fold changes were calculated using the fluorescence (a.u.) of pG3n5+pTrG3n5 at 10 mM IPTG divided by that of the following conditions: pG3n5+pTrG3n5 at 0 mM IPTG (~20X); pG3n5 only at 10 mM IPTG (~110X); and pG3n5 only at 0 mM IPTG (~270X). The error bars represent the standard deviation of the values from three biological replicates.

Supplementary Table S1. Plasmids used in this study.

S1-1. Plasmids used for *E. coli* DH10B strains.

Name	Parts	Figures
pG3n5	p15A ori; kan-R; P_{BBa_J23104}-switch G3n5; BBa B0012	Fig. 1A, 1B, Fig. S7
pG5-G3n5	p15A ori; kan-R; P_{BBa_J23104}-switch G5-G3n5; BBa B0012	Fig. 1C, 1D, Fig. S2A
pG5-G3n8	p15A ori; kan-R; P_{BBa_J23104}-switch G5-G3n8; BBa B0012	Fig. 1C, 1D, Fig. S2B
pG5-G3n11	p15A ori; kan-R; P_{BBa_J23104}-switch G5-G3n11; BBa B0012	Fig. 1C, 1D, Fig. S2C
pTrG3n5	ColE1 ori; amp-R; Plac-TrG3n5; his[min]	Fig. 1A, 1B, Fig. S7
pTrG5-TrG3n5	ColE1 ori; amp-R; Plac-TrG3n5; Plux-TrG5; his[min]	Fig. 1C, 1D, Fig. S2A
pTrG5-TrG3n8	ColE1 ori; amp-R; Plac-TrG3n8; Plux-TrG5; his[min]	Fig. 1C, 1D, Fig. S2B
pTrG5-TrG3n11	ColE1 ori; amp-R; Plac-TrG3n11; Plux-TrG5; his[min]	Fig. 1C, 1D, Fig. S2C
plac-gfpmut3	ColE1 ori; amp-R; Plac-gfpmut3; his[min]	Fig. S3A
plux-gfpmut3	ColE1 ori; amp-R; Plux-gfpmut3; his[min]	Fig. S3B
pT0-T3	p15A ori; kan-R; P_{BBa_J23104}-switch T0-T3; BBa B0012	Fig. 2B, Fig. S4
pT3	p15A ori; kan-R; P_{BBa_J23104}-switch T3; BBa B0012	Fig. 2B, Fig. S4
pTrT3	ColE1 ori; amp-R; Plac-TrT3; his[min]	Fig. 2B, Fig. S4

S1-2. Plasmids used for *E. coli* BL21 Star (DE3) strains.

Name	Parts	Figures
Pcon-sw2albion3-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion3; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion4-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion4; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion5-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion5; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion6-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion6; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion7-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion7; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion8-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion8; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion9-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion9; gfpmut3	Fig. 2C, Fig. S5

Pcon-sw2albion11-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion11; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion12-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion12; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion13-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion13; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion14-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion14; gfpmut3	Fig. 2C, Fig. S5
Pcon-sw2albion15-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch sw2albion15; gfpmut3	Fig. 2C, Fig. S5
Pcon-swVKFT3.B1-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch swVKFT3.B1; gfpmut3	Fig. 2D, Fig. S6
Pcon-swVKFT3.I2-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch swVKFT3.I2; gfpmut3	Fig. 2D, Fig. S6
Pcon-swVKFT3.V1 13-gfp-p15A	p15A ori; Kan-R; P_{BBa_J23104}-switch swVKFT3.V1-13; gfpmut3	Fig. 2D, Fig. S6
Plac-gfp-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-gfpmut3; sgl Term terminator	Fig. S3A
Plac-tr2alb3-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb3; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb4-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb4; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb5-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb5; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb6-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb6; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb7-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb7; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb8-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb8; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb9-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb9; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb11-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb11; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb12-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb12; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb13-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb13; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb14-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb14; sgl Term terminator	Fig. 2C, Fig. S5
Plac-tr2alb15-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger tr2alb15; sgl Term terminator	Fig. 2C, Fig. S5
Plac-trVKFT3-ColE1	ColE1; Amp-R; Lac Repressor; P_{LAC}-trigger trVKFT3; sgl Term terminator	Fig. 2D, Fig. S6

Supplementary Table S2. Genetic parts used in this study.

S2-1. Genetic parts for *E. coli* DH10B strains.

Type	Part Name	DNA sequence
Switch RNA	G3n5	AGCTATTACTACTTACCATTGTCTTGCTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
	G5-G3n5	TATGTAATTGATTTGGCTTCTGTTAGTTTCATACAAGAACTTAGACAATATGAAATCAACAGAAGCTATTACTACTTACCATTGTCTTGCTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
	G5-G3n8	TATGTAATTGATTTGGCTTCTGTTAGTTTCATACAAGAACTTAGACAATATGAAATCAACAGAAGCTACTACTTACCATTGTCTTGCTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
	G5-G3n11	TATGTAATTGATTTGGCTTCTGTTAGTTTCATACAAGAACTTAGACAATATGAAATCAACAGAAGCTACTTACCATTGTCTTGCTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
	T3	TCTAGACAAATCCGACCATTGTCTCACTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
	T0-T3	TCTAGACAAATCCGACCATTGTCTCACTCTCGCATTCCCGGTCTCGCGAGATCTAGACAAATCCGACCATTGTCTCACTCTATACAGAAAC**AGAGGAGA**TATAGA**ATG**AGACAATGG
Trigger RNA	TrG3n5	AGAGCAAGACAATGGTAAGTAGTAATAGCT
	TrG3n8	AGAGCAAGACAATGGTAAGTAGTAGCTTCT
	TrG3n11	AGAGCAAGACAATGGTAAGTAGCTTCTGTT
	TrG5[1]	GAAACTAACAGAAGCCAAATCAATTACATA
	TrT3	AGAGTGAGACAATGGTCGGATTTGTCTAGA
Promoter	BBa_J23104 (http://parts.igem.org/Part:BBa_J23104)	TTGACAGCTAGCTCAGTCCTAGGTATTGTGCTAGC
	P_{Lac}[6]	ATAAATGTGAGCGGATAACATTGACATTGTGAGCGGATAACAAGATACTGAGC
	P_{Lux}[6]	ACCTGTAGGATCGTACAGGTTTACGCAAGAAAATGGTTTGTTATAGTCGAATAAA
Terminator	BBa B0012[7]	TCACACTGGCTCACCTTCGGGTGGGCCTTTCTGCGTTTATA
	his [min] (S)[8]	GCCCCCGGAAGATCATTCCGGGGGCTTTTTTATT

Linker	21nt linker[1]	AACCTGGCGGCAGCGCAAAAG

Type	Part Name	DNA sequence
Gene	*gfpmut3*	ATGAGTAAAGGAGAAGAACTTTTCACTGGAGTTGT CCCAATTCTTGTTGAATTAGATGGTGATGTTAATGG GCACAAATTTTCTGTCAGTGGAGAGGGTGAAGGT GATGCAACATACGGAAAACTTACCCTTAAATTTATT TGCACTACTGGAAAACTACCTGTTCCATGGCCAAC ACTTGTCACTACTTTGACTTATGGTGTTCAATGCTT TTCAAGATACCCAGATCATATGAAACGGCATGACTT TTTCAAGAGTGCCATGCCCGAAGGTTATGTACAGG AAAGAACTATATTTTTCAAAGATGACGGGAACTAT AAGACACGTGCTGAAGTCAAGTTTGAAGGTGATA CACTTGTTAATAGAATCGAGTTAAAAGGTATTGATT TTAAAGAAGATGGAAACATTCTTGGACACAAGTTG GAATACAACTATAACTCACACAATGTATACATCATG GCAGACAAACAAAAGAATGGAATCAAAGTTAACT TCAAAATTAGACACAACATTGAAGATGGAAGCGTT CAACTAGCAGACCATTATCAACAAAATACTCCAATT GGCGATGGCCCTGTCCTTTTACCAGACAACCATTA CCTGTCCACACAATCTGCCCTTTCGAAAGATCCCA ACGAAAAGAGAGACCACATGGTCCTTCTTGAGTTT GTAACAGCTGCTGGGATTACACATGGCATGGATGA ACTATACAAAAGGCCTGCAGCAAACGACGAAAAC TACGCTTAAGTAGCTTAA

Sky blue: TrG3n* binding site (*=5, 8, or 11)

Pink: TrG5 binding site

Green: TrT3 binding site

Black bold: Ribosome binding site

Red bold: Start codon

S2-2. Genetic parts for *E. coli* BL21 Star (DE3) strains.

Switch RNA	Name on storage tubes	DNA sequence
Sw2v1	Sw2Albion3	ACAGCTCTATCCTGAACGATAGAGCCCTAGTTTCGAAACTTTACGTCGTGTTAC CTGTTATCTGGTAACACGAAAGCAAGTTTCGAAACTTTACGTCGTGTTACCTGT ATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGC GCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG

Sw2v2	Sw2Albion4	ACAGCTCTATCCTGAACGATAGAGCCCTATCTAGCTAGATTTACGTCGTGTTACCTGTCGATCGGTAACACGAAACTAATCTAGCTAGATTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v3	Sw2Albion5	ACAGCTCTATCCTGAACGATAGAGCCCTACTAATGATCATTTACGTCGTGTTACCTGTTATTCGGTAACACGAAACTAATGATCATTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v4	Sw2Albion6	ACAGCTCTATCCTGAACGATAGAGCCCTTAATGCATGCATTTACGTCGTGTTACCTGTATATCGGTAACACGAAACTAATGCATGCATTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v5	Sw2Albion7	ACAGCTCTATCCTGAACGATAGAGCCCTATTAGCTAATCTTTACGTCGTGTTACCTGTCATCCGGTAACACGAATCCAAGATTAGCTAATCTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v6	Sw2Albion8	ACAGCTCTATCCTGAACGATAGAGCCCTATCCGGATCTGTTTACGTCGTGTTACCTGTACTCCGGTAACACGAATCTAACAGATCCGGATCTGTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v7	Sw2Albion9	ACAGCTCTATCCTGAACGATAGAGCCCTGAATTCTTATGTTTACGTCGTGTTACCTGTATGTCGGTAACACGAAAGCAACATAAGAATTCTTATGTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v8	Sw2Albion11	ACAGCTCTATCCTGAACGATAGAGCCCTGCTAATCCGGATTTACGTCGTGTTACCTGTGCTCCGGTAACACGAAGCTAATCCGGATTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v9	Sw2Albion12	ACAGCTCTATCCTGAACGATAGAGCCCTGAACTTCGAAGTTTACGTCGTGTTACCTGTGTATTGGTAACACGAAACGAACTTCGAAGTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v10	Sw2Albion13	ACAGCTCTATCCTGAACGATAGAGCCCTATTCTAGAATCTTTACGTCGTGTTACCTGTATTCCGGTAACACGAGACTAAGATTCTAGAATCTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v11	Sw2Albion14	ACAGCTCTATCCTGAACGATAGAGCCCTAAGATCTTTGCTTTACGTCGTGTTAC

		CTGTATTCCGGTAACACGAGACTAAGCAAAGATCTTTGCTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Sw2v12	Sw2Albion15	ACAGCTCTATCCTGAACGATAGAGCCCTGAATTCTATGCTTTACGTCGTGTTACCTGTGACCCGGTAACACGAATCCAAGCATAGAATTCTATGCTTTACGTCGTGTTACCTGTATACAAGAAC**AGAGGAGA**TATACAATGAACACGACGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Switch with 1 Binding Site	VKFT3.B1	GGGCTCCCAGTGGCAGGACTGGGAGTATGGGCTAGAAATCAGAATGGCGATGATTTAGTGCCATACAAGAAC**AGAGGAGA**TATGGCATGAAATCATCGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Switch with 2 Binding Sites	VKFT3.I2	GGGCTCCCAGTGGCAGGACTGGGAGTATGTCTGATTTACCAATTAGAAATCAGAATGGCGATGATTTAGTGCCTCCATGAATCGAACCGGAGGCTAGAAATCAGAATGGCGATGATTTAGTGCCATACAAGAAC**AGAGGAGA**TATGGCATGAAATCATCGAACCTGGCGGCAGCGCAAAAGATGAGTAAAGGAGAAGAACTTTTCACTGG
Switch with 3 Binding Sites	VKFT3.V1	GGGCTCCCAGTGGCAGGACTGGGAGTATGTCTGATTTCAACCAATTAGAAATCAGAATGGCGATGATTTAGTGCCGGTCACCTATAGTACGCACCTACCGGCTAGAAATCAGAATGGCGATGATTTAGTGCCTCCATGAATCGAACCGGAGGCTAGAAATCAGAATGGCGATGATTTAGTGCCATACAAGAAC**AGAGGAGA**TATGGCATGAAATCATCGAACCTGGCGGCAGCGCAAAAG ATGAGTAAAGGAGAAGAACTTTTCACTGG

Trigger RNA for	Name on storage tubes	DNA sequence
Sw2v1	Tr2Albion3	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAAGTTTCGAAACTCCG
Sw2v2	Tr2Albion4	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAATCTAGCTAGATCCG
Sw2v3	Tr2Albion5	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAATGATCATTAGTCCG
Sw2v4	Tr2Albion6	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAATGCATGCATTACCG
Sw2v5	Tr2Albion7	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAAGATTAGCTAATCCG
Sw2v6	Tr2Albion8	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAACAGATCCGGATCCG
Sw2v7	Tr2Albion9	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAACATAAGAATTCCCG
Sw2v8	Tr2Albion11	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAATCCGGATTAGCCCG
Sw2v9	Tr2Albion12	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAACTTCGAAGTTCCCG
Sw2v10	Tr2Albion13	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAAGATTCTAG

		AATCCG
Sw2v11	Tr2Albion14	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAAGCAAAGATCTTCCG
Sw2v12	Tr2Albion15	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTACAGGTAACACGACGTAAAGCATAGAATTCCCG
Switch with 1-3 Binding Sites	TrVKFT3	CTACCGACAGGCCTCGAATGGCCTGTCGGCCTGGCACTAAATCATCGCCATTCTGATTTCTACCG
sgl Term terminator		CCAGGCATCAAATAAAACGAAAGGCTCAGTCGAAAGACTGGGCCTTTCGTTTTATCTGTTGTTTGTCG

Blue = Binding site for trigger RNA
Red = Start codon
Green = binding sequence
Black Bold = Ribosome binding site

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